

FinSherlock AI

Autonomous Agentic AML & Financial Crime Investigation Platform

Dynamic Intent Parsing · Zero-Hallucination Execution · IBM AML Dataset Integration

TEAM HEISENBERG

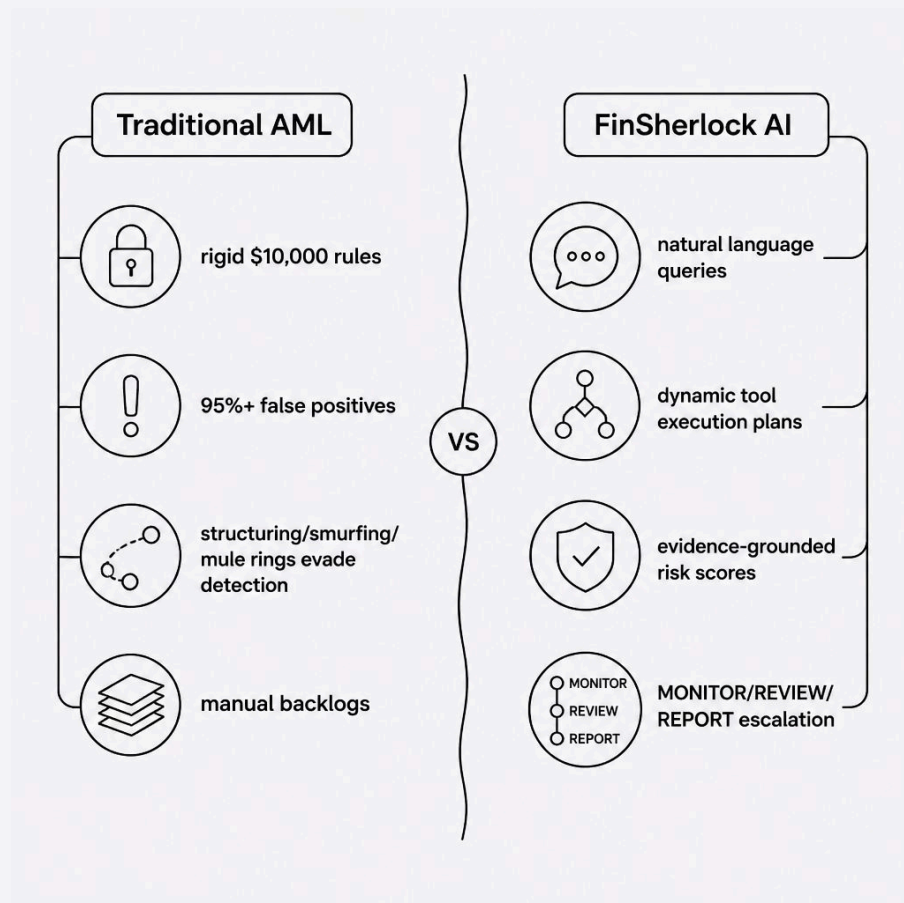
Solving High False-Positives in Anti-Money Laundering Compliance

Github Link: <https://github.com/Aashik1701/FinSherlock-AI>

Srilekha R · Mohammed Aashik F

Two Brain Safety Model · Zero-Hallucination Compliance Engine

95% False Positives - A System Built to Fail



What We Built

FinSherlock AI is an autonomous agentic investigator that accepts natural language queries, dynamically constructs custom tool execution plans, and generates evidence-grounded risk assessments with actionable escalation recommendations.

- IBM HI-Small Dataset: 500K+ transactions with ground-truth labels
- DuckDB in-memory engine for sub-second SQL execution
- Brain 1 (LLM Planner): intent parsing, dynamic JSON tool plans
- Brain 2 (Deterministic Engine): Python, NetworkX, XGBoost - zero hallucinations

Five AML Typology Engines



Structuring Detector

31 U.S.C. § 5324 - Detects sub-threshold cash clustering (\$9,500 - \$9,900) across branches.



Smurfing Fan-Out

Graph degree aggregation detecting multi-account fund dispersal patterns.



Layering Chain Detector

DFS pathfinding tracing rapid multi-hop transfers across accounts.



Money Mule Ring Finder

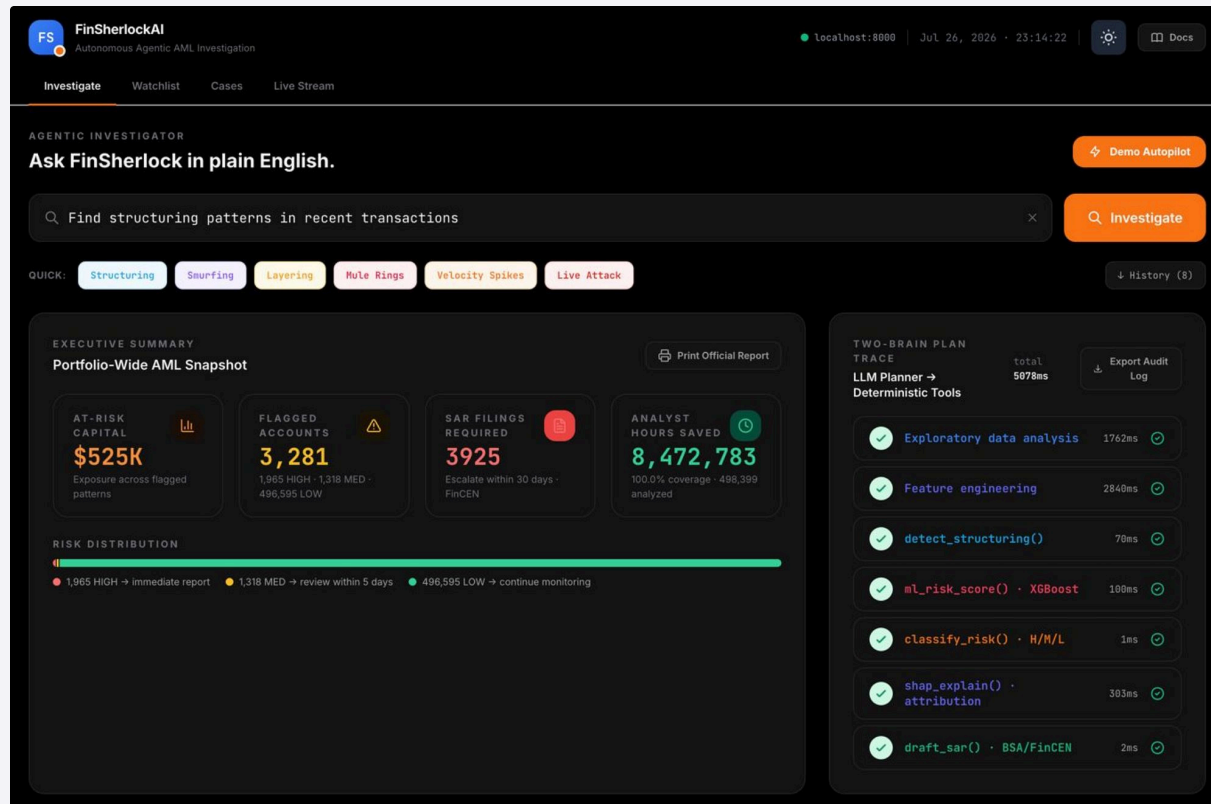
Louvain Community Detection uncovering closed fund-circulating clusters (modularity 0.71).



Velocity Surge Detector

Compares 7 day frequency against 90 day baselines to catch 3× behavioral spikes.

Auditability by Design - Built for Regulators



The Impact: \$25.1M saved per 500K transactions - 99.9% false-positive elimination.

Explainable AI (XAI)

- **SHAP Feature Attribution** - Horizontal waterfall charts showing exact positive/negative risk drivers
- **Counterfactual Reasoning** - Explains what behavior change would unflag an account
- **Digital Chain of Custody** - 1-click Export Audit Log JSON with execution plans, latencies, and parameters

What Sets Us Apart

- **1 Click SAR Export** - Pre-filled FinCEN Suspicious Activity Report (BSA Form 111) draft
- **Active Learning Loop** - Analyst feedback triggers model retraining endpoints
- **\$25.1M Saved** per 500K transactions - eliminates 99.9% of false-positive overhead

SUMMARY

FinSherlock AI - Key Takeaways

The Problem

95%+ false positives from rigid rule-based AML systems

The Architecture

Two-Brain Agentic System: LLM Planner + Deterministic Engine

The Coverage

5 typology engines covering structuring, smurfing, layering, mules, velocity

The Impact

\$25.1M saved per 500K transactions - 99.9% false positive elimination

Problem Statement 1

TEAM HEISENBERG · Srilekha R · Mohammed Aashik F · *Société Générale Hackathon*

